ChatGPT

L20.pdf
PDF

Create a lecture scribe for lecture 20 that is given and it should be for revision purpose during the exam please note that all the formulas and important topic is covered in scribe. Generate a latex code scribe in which the content is strictly from ppt only. Don't add any additional steps from you. Make it in a concise, clear and simple to study and understand but to make it simple don't add any additional simple versions or steps. All theorems and proofs should be mentioned clearly using /section of required use /subsection. If there is any matrix given then also explain that properly step by step as per ppt.The scribe should contain course name i.e. CSE400 FUNDAMENTALS OF PROBABILITY IN COMPUTING, date of lecture and all the necessary details. All this content should be under one file starting from \documentclass[11pt]{article} to end with \end{document}. Give me the latex code.

LaTeX

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\documentclass[11pt]{article}
\usepackage{amsmath, amssymb}

\title{CSE400: Fundamentals of Probability in Computing}
\author{Lecture 20: Linear Transformations and Expectations of Multiple RVs}
\date{March 19, 2025}

\begin{document}

\maketitle

\section{Outline}
\begin{itemize}
    \item Statistical Representation of Random Vectors
    \item Mean Vector, Correlation, and Covariance Matrices
    \item Linear Transformations
    \item Mapping properties under $Y = AX + b$
    \item Multivariate Gaussian Distributions
    \item From 1D/2D cases to N-dimensional Joint PDFs
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    \item Special Cases and Properties
    \item Implications of zero correlation and matrix factorization
\end{itemize}

\section{Statistical Representation of Random Vectors}
\subsection{Definition}
Let $\mathbf{X}$ be an $N$-dimensional random vector:
\[
\mathbf{X} =
\begin{bmatrix}
X_1 \\
X_2 \\
\vdots \\
X_N
\end{bmatrix}
\]

\section{Mean Vector}
\subsection{Definition}
The mean vector is:
\[
\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}] =
\begin{bmatrix}
\mathbb{E}[X_1] \\
\mathbb{E}[X_2] \\
\vdots \\
\mathbb{E}[X_N]
\end{bmatrix}
\]

\section{Covariance Matrix}
\subsection{Definition}
The covariance matrix is:
\[
\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu}
\]

\subsection{Interpretation}
\begin{itemize}
    \item Represents dependencies between variables
    \item Acts as a “dependency map”
\end{itemize}

\section{Multivariate Gaussian Distribution}
\subsection{Assumption}
\[
\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)
\]

\subsection{Implication}
The system is defined by:
\begin{itemize}
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    \item Mean (typical conditions)
    \item Covariance (interdependencies)
\end{itemize}

\section{Linear Transformations}
\subsection{Definition}
\[
\mathbf{Y} = A\mathbf{X} + \mathbf{b}
\]

\subsection{Mapping Properties}
\begin{itemize}
    \item Linear transformation of random vector
\end{itemize}

\section{Case Study: Smart City Traffic + Weather System}

\subsection{Step 1: Define Random Variables}
\[
\mathbf{X} =
\begin{bmatrix}
X_1 \\
X_2 \\
X_3 \\
X_4 \\
X_5
\end{bmatrix}
\]
(N-dimensional random vector, here $N = 5$)

\subsection{Step 2: Mean Vector}
\[
\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]
\]

\subsection{Step 3: Covariance Matrix Interpretation}
\begin{itemize}
    \item More rain $\Rightarrow$ more congestion
    \item More rain $\Rightarrow$ lower speed
    \item More traffic $\Rightarrow$ higher pollution
    \item Higher temperature $\Rightarrow$ pollution disperses
\end{itemize}

\subsection{Step 4: Joint Gaussian Distribution}
\[
\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)
\]

\subsection{Step 6: Transformation}
Define congestion index:
\[
\mathbf{Y} = A\mathbf{X} + \mathbf{b}
```

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\]

\subsection{Inference}
If rainfall increases:
\begin{itemize}
    \item Speed $\downarrow$
    \item Traffic $\uparrow$
    \item Pollution $\uparrow$
\end{itemize}

\end{document}
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